SESSION 6

Hackathon Readiness — The GeoAI Ethics Checklist

Seven Judging Dimensions · Human Oversight in Depth · Ethics Audit Exercise · Presenting Ethics in Your Pitch · Module Wrap-Up · Resources

Day 1 · 7 July 2026 · GeoAI Hackathon Programme

How to use these notes. *These notes are self-contained: the full seven-dimension checklist with judge-level detail, an expanded explanation of Human Oversight, the audit exercise, guidance on presenting ethics in your final pitch, and the wrap-up of the whole FD1001 module.*

Learning Objectives

By the end of this session, you should be able to:

- Name the seven dimensions of the GeoAI Hackathon Ethics Checklist against which every project is judged.
- Explain Human Oversight in depth, including its basis in international AI governance.
- Run a traffic-light self-audit of any AI system against all seven dimensions (Activity 6).
- Present your project's ethics work convincingly in a hackathon pitch.
- Summarise the five key takeaways of the full FD1001 module.

1. The GeoAI Hackathon Ethics Checklist

Every dimension below is part of how your hackathon project will be evaluated — judges score against these same seven dimensions. Use the checklist as a self-audit before every submission.

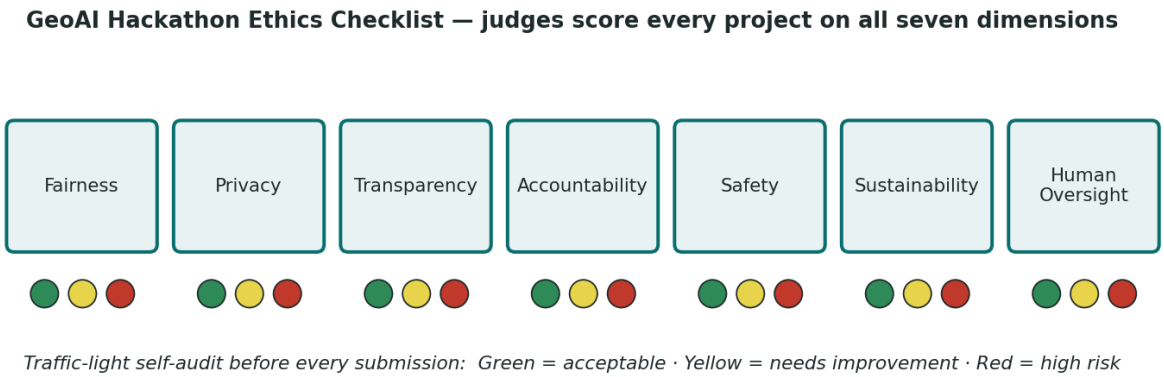


Figure 6.1 — The seven-dimension checklist with its traffic-light scoring approach.

Dimension	What judges are looking for
Fairness	The system works about as well for every group it touches — no group is systematically helped or harmed more than another. Evidence: accuracy checked per group/area, not just overall (Session 4, Principle 1).
Privacy	Location and personal data are collected with knowledge and consent, at the coarsest resolution the task allows, with re-identification risk addressed via aggregation, k-anonymity, or differential privacy (Sessions 3–4).

Dimension	What judges are looking for
Transparency	Affected people can understand, in plain terms, roughly how and why a decision was reached — e.g. alerts that show their top driver, like Team Tropika's "Based on: rainfall in your area" (Sessions 4–5).
Accountability	A named person owns the system's decisions and is answerable when it causes harm (Session 4, Principle 4; Team Tropika Step 4).
Safety	The system performs consistently under real conditions and fails loudly and safely — able to say "I'm not sure" via confidence flags — rather than confidently wrong (Session 4, Principle 5).
Sustainability	The environmental and social cost of building and running the model is justified by the value delivered; smaller and well-tuned beats giant by default (Sessions 4–5).
Human Oversight	A human can review, override, and appeal the system's outputs before real-world action is taken on them — no fully automatic harm (expanded below).

2. Human Oversight, In Depth

Human Oversight is the seventh dimension — the one that extends the six Responsible AI principles from Session 4. It comes directly from Session 3's case-study rule ("pair any risk score with human review and a clear appeal path") and is now written into international law: Article 14 of the EU AI Act requires high-risk AI systems to be designed so natural persons can effectively oversee them while in use. In practice, effective human oversight means the humans in charge can:

- Understand the system's real capabilities and limitations — and monitor it well enough to spot anomalies and unexpected behaviour.
- Guard against automation bias — the documented tendency to over-trust and over-rely on machine output, especially under time pressure.
- Correctly interpret the system's output, given its confidence and known blind spots.
- Decide not to use the output in a given situation — and intervene or safely stop the system (a "stop button" or equivalent).

For your hackathon project, oversight questions to answer explicitly: Who sees a prediction before action is taken on it? Can they override it? How does an affected person appeal or report a wrong output? Team Tropika's "Reply MISSED if you flooded without an alert" is a minimal but real oversight channel.

3. Hands-On Activity 6 — Ethics Audit Exercise (10 minutes)

Task: evaluate an AI-based flood warning system against each of the seven dimensions using the traffic-light approach:

- **Green** = Acceptable — evidence exists that this dimension is handled.
- **Yellow** = Needs improvement — partially handled, or handled without evidence.
- **Red** = High risk — unaddressed, and could cause real harm.

Dimension	Green	Yellow	Red	Notes
Fairness				
Privacy				
Transparency				

Dimension	Green	Yellow	Red	Notes
Accountability				
Safety				
Sustainability				
Human Oversight				

Tip: for every Yellow or Red, write one concrete mitigation in the Notes column — this table then becomes the seed of your project's Ethical Risk Register (Session 5, Activity 5).

Worked hints for the flood-warning scenario: Fairness — is accuracy tested in low-gauge areas (Yellow if unknown)? Privacy — does it store exact household coordinates it doesn't need (Red if yes)? Transparency — do alerts explain themselves (Yellow if just a score)? Accountability — is anyone named (Red if no)? Safety — does it show confidence (Yellow if silent)? Sustainability — is the model right-sized (usually Green for a small model)? Human Oversight — can residents report a missed flood (Green if yes)?

4. Presenting Ethics in Your Final Pitch

Judges consistently reward teams that treat ethics as engineering, not apology. A strong 60-second ethics segment in a pitch contains four artefacts — all of which this module has already taught you to build:

1. **The risk register (Session 5, Activity 5):** "We identified five risks specific to our system — here are the top two."
2. **The priority grid (Session 5, Step 2):** "We scored them by likelihood and severity, which is why we spent our hours on these."
3. **The mitigation log (Session 5, Step 3):** "Here is what we built in — confidence tags, opt-in consent, plain-language explanations."
4. **The monitoring plan (Session 5, Step 4):** "And here is what happens after demo day — the metric, the checkpoint, the owner."

Bonus signal: reference the frameworks by name — "our design follows Malaysia's AIGE seven principles and would satisfy the human-oversight expectations of the EU AI Act" (Session 4). Few hackathon teams can say this; the ones who can, stand out.

5. Module Wrap-Up — Key Takeaways of FD1001

- **GeoAI blends AI with location data** to detect patterns, predict risk, and guide real decisions at scale (Session 1).
- **Innovation is a team sport:** Design Thinking and diverse stakeholder input make GeoAI solutions stronger (Session 2).
- **Every benefit carries a risk:** bias, privacy, explainability, and environmental cost all need conscious trade-offs (Session 3).
- **Six principles guide responsible AI:** fairness, privacy, transparency, accountability, safety, and sustainability — plus human oversight as the seventh judged dimension (Sessions 4 & 6).
- **Ethics is a design input, not an afterthought:** use the GeoAI Canvas and the Ethics Checklist from Day 1 onward (Sessions 1, 5 & 6).

6. Resources & Further Reading

GeoAI Foundations

- Esri — What is GeoAI? (esri.com/en-us/capabilities/geoai/overview).
- Picterra — GeoAI Explained (2025).
- Harvard CGA — Critical Challenges of GeoAI.

AI Ethics & Responsible AI

- ISO — Building a Responsible AI; Microsoft Responsible AI Standard; Harvard DCE — 5 Key Principles of Responsible AI.
- OECD — AI Principles; EU AI Act (artificialintelligenceact.eu), especially Article 14 on Human Oversight.
- MOSTI — National Guidelines on AI Governance & Ethics (AIGE, Sept 2024); ASEAN Guide on AI Governance and Ethics (Feb 2024).

Design Thinking & Innovation

- Stanford d.school — Design Thinking Bootleg and Process Guide; IDEO.org — Design Kit; IDEO — Human-Centred Design Toolkit.

Full citations are available on request from the training team — see the facilitators for direct links.

Self-Check Questions

1. List the seven checklist dimensions from memory, then check yourself against Section 1.
2. What is automation bias, and which checklist dimension is designed to counter it?
3. A team's flood app has no way for a resident to report a missed alert. Which dimension(s) go Yellow or Red, and what one-line fix would you propose?
4. Name the four ethics artefacts a strong pitch presents, and which session's activity produces each.
5. Why does the checklist have seven dimensions when Session 4 taught six principles? Where does the seventh come from?

What Happens Next

After the mid-day break (10.45 – 11.00), the next module covers Team Formation & Problem Scoping for your GeoAI Hackathon project. Bring your Activity 1 Problem–Data–Solution Canvas, your Activity 2 problem definition, and your Activity 6 ethics audit — they are the raw material for your team's project pitch.